

The effects of gravity are effectively negated if an object is in free fall. For example, an object orbiting the Earth experiences an apparent weightlessness known as microgravity. Because the human body evolved under the influence of Earth's gravity, human physiology undergoes several changes in microgravity environments. Puffy face syndrome (PFS) occurs when an astronaut's extracellular fluid shifts toward upper body regions, resulting in facial bulging and leg shrinkage. Elongation of the spinal column also occurs as intervertebral discs (spinal cartilage) decompress, making astronauts up to 6 cm taller.

Researchers simulated microgravity-induced spinal elongation on Earth by suspending 50 volunteers by the arms, holding each above the ground in an upright position. The thickness of each intervertebral disc was measured prior to and following 10 minutes of suspension. Researchers then compared the percentage change in intervertebral disc thickness to similar measurements recorded in microgravity, as shown in Figure 1.

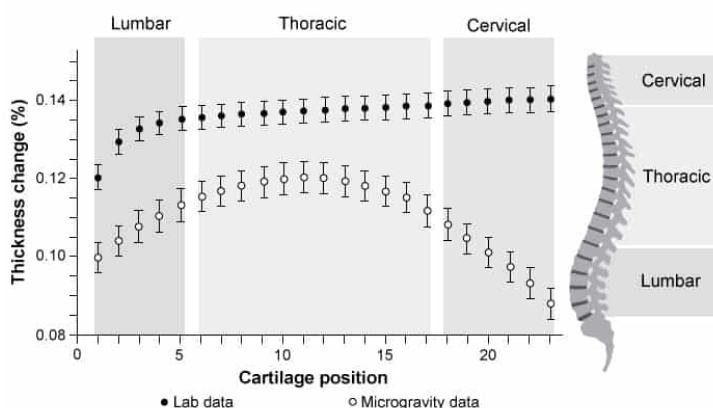


Figure 1 Percentage change in intervertebral disc thickness

In theory, spinal elongation would also occur in reduced-gravity environments like the surface of the Moon, where the acceleration of gravity is one-sixth that observed on the Earth's surface.

A volunteer with a mass of 80 kg is suspended by his arms on Earth. What is the approximate tension on lumbar disc 5, which is located

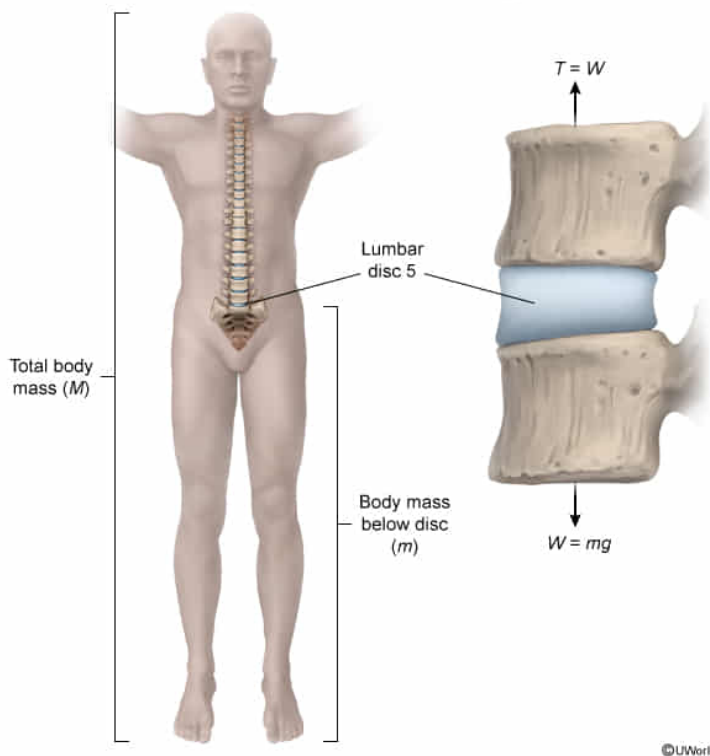
near the volunteer's center of mass? (Note: The acceleration of gravity is $g = 10 \text{ m/s}^2$)

- ☐ A. 40 N [6%]
- ☐ B. 70 N [12%]
- ✓ ☒ C. 400 N [45%]
- ☐ D. 700 N [35%]

Correct

45% answered correctly

Explanation:



Newton's second law of motion dictates that an object located in three-dimensional space will **accelerate** when the sum of all forces acting along one or more axes of the xyz-coordinate system does not equal zero. When the **vector components** of each force acting on the object are known, **free-body diagrams** can be used to assess whether an object accelerates or remains in equilibrium (ie, unchanging zero or non-zero velocity).

While suspended, the volunteer's intervertebral discs must support the

weight of the body below it for the spine to remain motionless in **static equilibrium** (ie, unchanging zero velocity). **Weight** W is the gravitational force acting to pull the suspended volunteer downward (-y direction), and the magnitude of weight is a product of the object's mass m and the acceleration of gravity g :

$$W = mg$$

Because the spine of a suspended volunteer does not accelerate, the weight of the body below the discs must be negated by an antagonistic upward force (+y direction). Furthermore, the volunteer is suspended above the ground, so the force opposing gravity must be the **tension** T experienced by the discs and other tissues.

Lumbar disc 5 is located close to the volunteer's center of mass. Therefore, approximately half the volunteer's body mass is located below lumbar disc 5. As a result, T on lumbar disc 5 is equal to approximately half the volunteer's W :

$$T = \frac{Mg}{2} = \frac{(80 \text{ kg})(10 \text{ m/s}^2)}{2} = 400 \text{ N}$$

(Choice A) Half of the volunteer's total mass is 40 kg, a value that is multiplied by the acceleration of gravity (10 m/s^2) to obtain weight.

(Choice B) 70 N is obtained by incorrectly calculating 90% of the volunteer's mass, instead of half of the volunteer's weight.

(Choice D) 700 N is obtained by incorrectly calculating 90% of the volunteer's weight, instead of half of the volunteer's weight.

Educational objective:

Newton's second law dictates that an object will accelerate unless the forces acting on the object completely negate one another. For objects suspended in air, the weight of the object is typically countered by a tension force that maintains static equilibrium.

Subject:
Physics

System:
Mechanics and Energy

The effects of gravity are effectively negated if an object is in free fall. For example, an object orbiting the Earth experiences an apparent weightlessness known as microgravity. Because the human body evolved under the influence of Earth's gravity, human physiology undergoes several changes in microgravity environments. Puffy face syndrome (PFS) occurs when an astronaut's extracellular fluid shifts toward upper body regions, resulting in facial bulging and leg shrinkage. Elongation of the spinal column also occurs as intervertebral discs (spinal cartilage) decompress, making astronauts up to 6 cm taller.

Researchers simulated microgravity-induced spinal elongation on Earth by suspending 50 volunteers by the arms, holding each above the ground in an upright position. The thickness of each intervertebral disc was measured prior to and following 10 minutes of suspension. Researchers then compared the percentage change in intervertebral disc thickness to similar measurements recorded in microgravity, as shown in Figure 1.

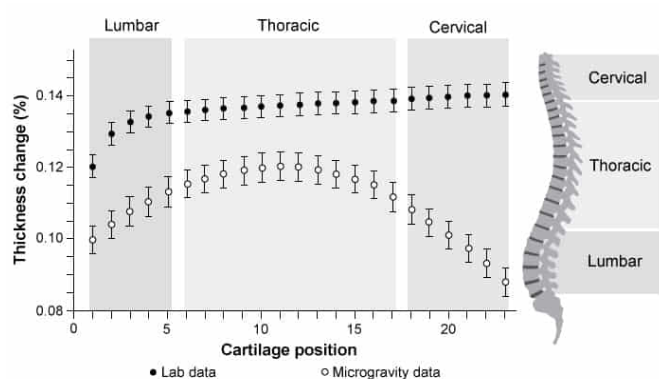


Figure 1 Percentage change in intervertebral disc thickness

In theory, spinal elongation would also occur in reduced-gravity environments like the surface of the Moon, where the acceleration of gravity is one-sixth that observed on the Earth's surface.

According to the passage, which of the following changes occurs as an astronaut leaves Earth's surface and enters orbit?

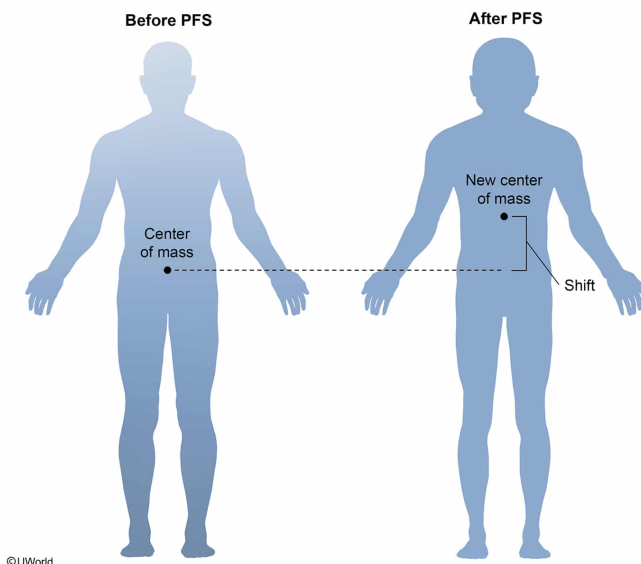
- ☐ A. Coefficient of kinetic friction between the astronaut and nearby objects. [4%]
- ☒ B. Astronaut center of mass. [41%]
- ☐ C. Astronaut bodily inertia. [25%]

- ☐ D. Magnitude of the gravitational force between the astronaut and the spacecraft. [27%]

Correct

42% answered correctly

Explanation:



The **center of mass** (CM) of a **system of point masses** is the average of the masses m_i weighted by their displacement r_i from a fixed reference point:

$$r_{\text{CM}} = \frac{m_1 r_1 + m_2 r_2 + m_3 r_3 + \dots}{m_1 + m_2 + m_3 + \dots}$$

In the case of a body with a non-uniform mass distribution (eg, humans), the CM is closer to the largest collection or concentration of mass. Furthermore, if a **shift in mass** occurs in one direction (ie, r_i increases or decreases), the **CM will also shift** in the same direction.

According to the passage, an orbit around Earth is a microgravity environment. In microgravity, PFS is observed as extracellular fluid previously contained within lower portions of the body (ie, feet and legs) shifts to the upper portions of the body (ie, chest and head). Consequently, the shift in an individual's fluid mass will result in a shift in the individual's center of mass.

(Choice A) The coefficient of kinetic friction depends only on the properties of the two surfaces that are in contact. The friction force may change due to the effects of gravity on the normal forces between two objects, but the coefficient of friction remains constant.

(Choice C) Inertia refers to an object's tendency to oppose changes in motion. Inertia depends only on the characteristics of the object itself (ie, mass), not on external factors such as gravity or position.

(Choice D) The mass of the astronaut will not change upon entering low-earth orbit. Consequently, the magnitude of the **gravitational force** between the astronaut and nearby objects (eg, the spacecraft) will remain the same.

Educational objective:

The center of mass of a system of masses is the average of the masses weighted by their displacement from a fixed reference point. The center of mass shifts in the same direction as a redistribution of mass.

Time spent:0 sec

Subject:
Physics

QID:400626

System:
Mechanics and Energy

The effects of gravity are effectively negated if an object is in free fall. For example, an object orbiting the Earth experiences an apparent weightlessness known as microgravity. Because the human body evolved under the influence of Earth's gravity, human physiology undergoes several changes in microgravity environments. Puffy face syndrome (PFS) occurs when an astronaut's extracellular fluid shifts toward upper body regions, resulting in facial bulging and leg shrinkage. Elongation of the spinal column also occurs as intervertebral discs (spinal cartilage) decompress, making astronauts up to 6 cm taller.

Researchers simulated microgravity-induced spinal elongation on Earth by suspending 50 volunteers by the arms, holding each above the ground in an upright position. The thickness of each intervertebral disc was measured prior to and following 10 minutes of suspension. Researchers then compared the percentage change in intervertebral disc thickness to similar measurements recorded in microgravity, as shown in Figure 1.

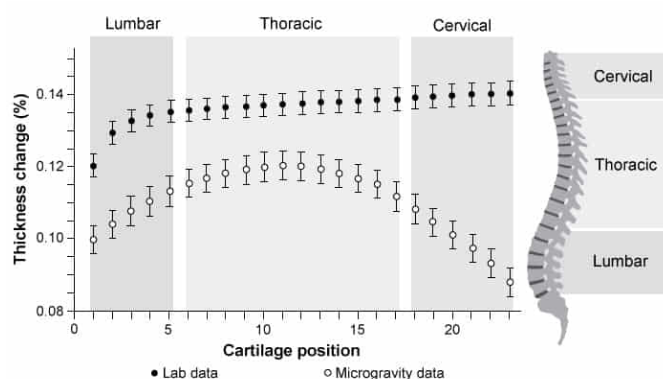


Figure 1 Percentage change in intervertebral disc thickness

In theory, spinal elongation would also occur in reduced-gravity environments like the surface of the Moon, where the acceleration of gravity is one-sixth that observed on the Earth's surface.

An astronaut drifts through space along the surface of her spaceship. She is perpendicular to the spaceship with her feet toward it. The astronaut moves as a rigid body with constant velocity and no rotation. Which of the following occurs when the astronaut momentarily strikes a protruding portion of the spaceship with her foot? The astronaut:

- ☐ A. slows down and continues along her original path with no rotation.

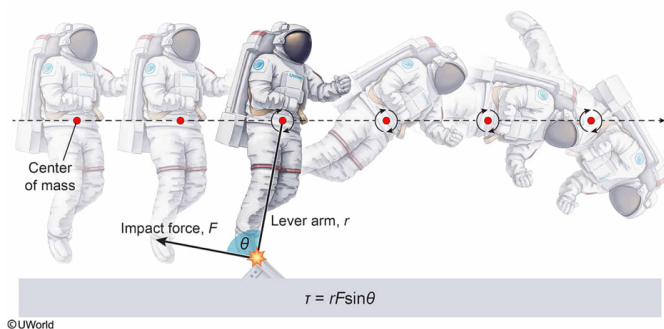
[12%]

- ☐ B. turns upside down and slowly falls to the ship's surface. [9%]
- ✓ ☒ C. continues along her original path while spinning about her center of mass. [71%]
- ☐ D. stops, spins about the protrusion, and falls to the ship. [6%]

Correct

72% answered correctly

Explanation:



A torque is a rotational force that makes an object spin about a pivot point. A pivot point can be a fixed point such as a hinge or axle seen in mechanical systems. For an unrestrained object like an astronaut floating in space, the **pivot point** is the **center of mass CM**.

Torque is generated by a force F applied perpendicularly to a lever arm (ie, the structure connected to the pivot point):

$$T = r \cdot F\sin(\theta)$$

where θ is the angle between the lever arm and the force, and r is the length of the lever arm.

Upon striking the protrusion of the ship, the astronaut's foot experiences an **impact force** due to the collision. The foot, leg, and torso act as the lever arm between the impact force and the astronaut's center of mass. Therefore, the collision **generates a torque** about the astronaut's CM, making the **astronaut spin**.

The astronaut's translational (ie, non-rotational) motion does not change significantly because **gravitational forces are negated** in microgravity. However, the astronaut continues along her **original path** with reduced velocity because a portion of the impact force acts perpendicular to her

original path, causing negative acceleration.

(Choice A) The impact between the astronaut's foot and the protrusion on the ship generates a torque, making her spin about her center of mass.

(Choices B and D) The astronaut does not fall to the ship because the effects of the gravitational force are minimal in microgravity. The impact with the protrusion induces rotation but the motion of her center of mass is mostly unchanged.

Educational objective:

Torque is a rotational force that makes objects rotate about a pivot point. For an unrestrained object, the pivot point is the object's center of mass.

Time spent:0 sec

Subject:
Physics

QID:400627

System:
Mechanics and Energy

The effects of gravity are effectively negated if an object is in free fall. For example, an object orbiting the Earth experiences an apparent weightlessness known as microgravity. Because the human body evolved under the influence of Earth's gravity, human physiology undergoes several changes in microgravity environments. Puffy face syndrome (PFS) occurs when an astronaut's extracellular fluid shifts toward upper body regions, resulting in facial bulging and leg shrinkage. Elongation of the spinal column also occurs as intervertebral discs (spinal cartilage) decompress, making astronauts up to 6 cm taller.

Researchers simulated microgravity-induced spinal elongation on Earth by suspending 50 volunteers by the arms, holding each above the ground in an upright position. The thickness of each intervertebral disc was measured prior to and following 10 minutes of suspension. Researchers then compared the percentage change in intervertebral disc thickness to similar measurements recorded in microgravity, as shown in Figure 1.

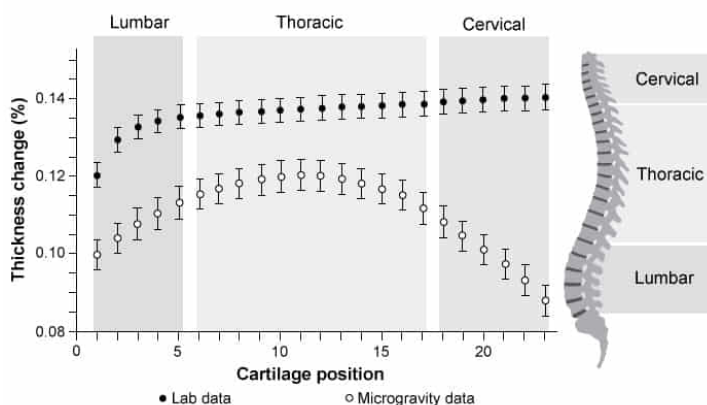


Figure 1 Percentage change in intervertebral disc thickness

In theory, spinal elongation would also occur in reduced-gravity environments like the surface of the Moon, where the acceleration of gravity is one-sixth that observed on the Earth's surface.

Ignoring any attachments between the spinal discs and vertebrae, how would the maximum force of static friction between these anatomical

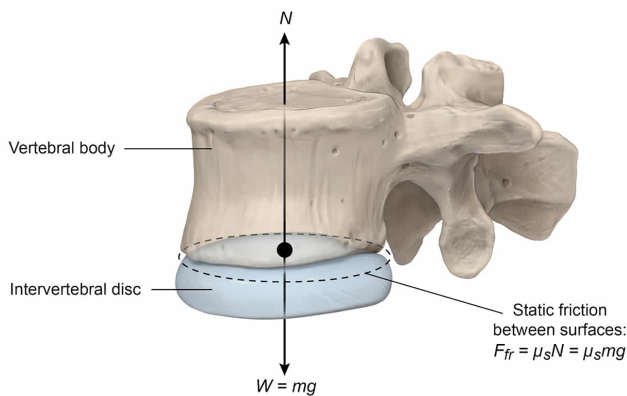
structures compare for a person standing on the surface of the Moon to a person standing on the surface of the Earth?

- ☐ A. It would be 6 times as great. [7%]
- ☐ B. It would be approximately equal. [6%]
- ✓ ☒ C. It would be 1/6 as great. [83%]
- ☐ D. It would be 0. [2%]

Correct

83% answered correctly

Explanation:



Friction refers to the force that resists sliding between two surfaces. Friction negates (partially or completely) any forces that promote sliding. For example, **static friction** F_{fr} occurs when the frictional force is equal and opposite to the combination of forces promoting motion. The magnitude of F_{fr} increases until the forces promoting motion exceed the maximum value of F_{fr} and the two surfaces start to **slide against each other**.

The maximum static friction force is equal to the product of the **coefficient of static friction** μ_s , and the **normal force** N , which is typically equal to the **weight** mg :

$$F_{fr} = \mu_s N = \mu_s mg$$

Within the spinal column, the N between the intervertebral discs and the

vertebrae is equal to the weight of the body above the discs.

Because both m and μ_s are intrinsic properties, they are not affected by situational factors such as gravity. Therefore, only a change in g will influence the maximum F_{fr} observed on the Moon relative to the maximum F_{fr} observed on the Earth.

Specifically, the maximum value of F_{fr} on the surface of the Moon will be 1/6 the maximum value of F_{fr} on the surface of the Earth because g on the Moon is 1/6 the magnitude of g on the Earth.

(Choices A and B) Because the force of gravity on the Moon is 1/6 the force of gravity on the Earth, the maximum static friction force on the Moon will be 1/6 as great the maximum force on the Earth, not 6 times as great or approximately equal.

(Choice D) The passage explains that the Moon is a reduced-gravity environment, *not* a microgravity environment. On the Moon, the effects of gravity are not eliminated.

Educational objective:

Friction is the force that resists sliding between two surfaces. Maximal static friction is proportional to the normal force exerted by the surfaces. Reducing gravitational acceleration also reduces the normal force, which in turn reduces the static friction by the same degree.

Time spent:0 sec

Subject:

Physics

QID:400628

System:

Mechanics and Energy

The effects of gravity are effectively negated if an object is in free fall. For example, an object orbiting the Earth experiences an apparent weightlessness known as microgravity. Because the human body evolved under the influence of Earth's gravity, human physiology undergoes several changes in microgravity environments. Puffy face syndrome (PFS) occurs when an astronaut's extracellular fluid shifts toward upper body regions, resulting in facial bulging and leg shrinkage. Elongation of the spinal column also occurs as intervertebral discs (spinal cartilage) decompress, making astronauts up to 6 cm taller.

Researchers simulated microgravity-induced spinal elongation on Earth by suspending 50 volunteers by the arms, holding each above the ground in an upright position. The thickness of each intervertebral disc was measured prior to and following 10 minutes of suspension. Researchers then compared the percentage change in intervertebral disc thickness to similar measurements recorded in microgravity, as shown in Figure 1.

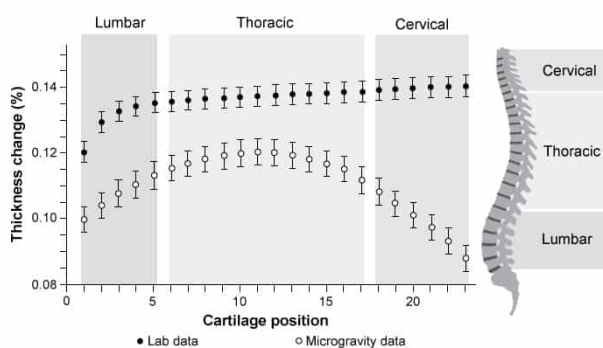
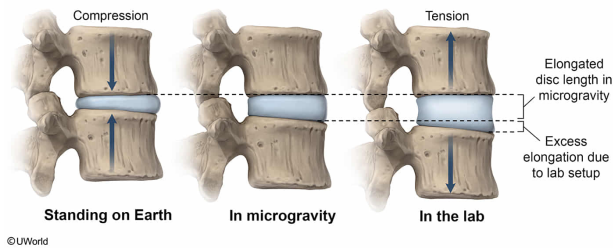


Figure 1 Percentage change in intervertebral disc thickness

In theory, spinal elongation would also occur in reduced-gravity environments like the surface of the Moon, where the acceleration of gravity is one-sixth that observed on the Earth's surface.

Which of the following statements explains the discrepancy between the microgravity data and the laboratory data for the experiment described in the passage?

- ☐ A. In microgravity, gravitational forces were converted to tension. [8%]
- ✓ ☒ B. In the laboratory setting, the weight of the body still acted on spinal cartilage. [72%]
- ☐ C. The mass of the laboratory volunteers did not decrease. [5%]
- ☐ D. Laboratory volunteers were not suspended upside-down to account for PFS. [12%]

Explanation:

For people standing on Earth, each intervertebral disc is compressed due to the **weight** of the body above it. In **microgravity**, the effects of gravity are largely eliminated such that spinal column discs do not support any weight. Consequently, microgravity environments promote cartilage **decompression** and **elongation**. Decompression and elongation of the spinal column also occurs with the suspension of experimental subjects on Earth because each disc no longer supports the weight of bodily structures located above.

For the spine to remain motionless, suspension also forces each disc to support the weight of inferior body structures. Support of inferior structures occurs through **tension**, a type of force that results from an object being pulled in two directions.

In contrast to compression, tension acts to elongate objects. As a result, the extra spinal column length seen in Figure 1 of the laboratory data may be explained by tension due to body weight. Therefore, it cannot be said that the effects of gravity were eliminated; gravity still acted on the body but was experienced as tension rather than compression.

(Choice A) The effect of tension was not observed in the microgravity condition because little gravity was present in space. Greatly reduced gravity leads to greatly reduced weight.

(Choice C) Mass is an intrinsic property of an object that depends only on the characteristics of the object and not on external factors (eg, local gravity).

(Choice D) Although hanging a subject upside-down may reproduce the redistribution of bodily fluids experienced in microgravity, fluid weight pooling at the head of an upside-down subject suspended by the feet would contribute to further spinal elongation.

Educational objective:

Weight can act as a compressive force or as a tensile (pulling) force. Tensile forces may lead to the elongation of objects under tension.

Subject:
Physics

System:
Mechanics and Energy

The effects of gravity are effectively negated if an object is in free fall. For example, an object orbiting the Earth experiences an apparent weightlessness known as microgravity. Because the human body evolved under the influence of Earth's gravity, human physiology undergoes several changes in microgravity environments. Puffy face syndrome (PFS) occurs when an astronaut's extracellular fluid shifts toward upper body regions, resulting in facial bulging and leg shrinkage. Elongation of the spinal column also occurs as intervertebral discs (spinal cartilage) decompress, making astronauts up to 6 cm taller.

Researchers simulated microgravity-induced spinal elongation on Earth by suspending 50 volunteers by the arms, holding each above the ground in an upright position. The thickness of each intervertebral disc was measured prior to and following 10 minutes of suspension. Researchers then compared the percentage change in intervertebral disc thickness to similar measurements recorded in microgravity, as shown in Figure 1.

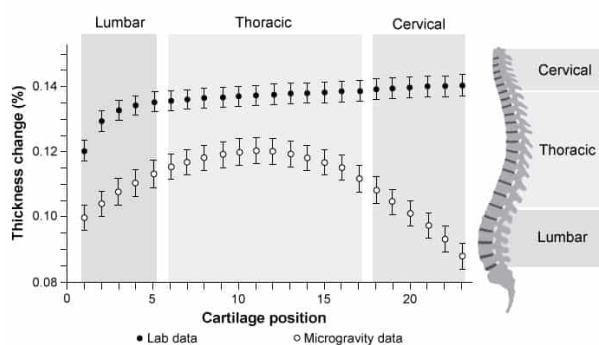


Figure 1 Percentage change in intervertebral disc thickness

In theory, spinal elongation would also occur in reduced-gravity environments like the surface of the Moon, where the acceleration of gravity is one-sixth that observed on the Earth's surface.

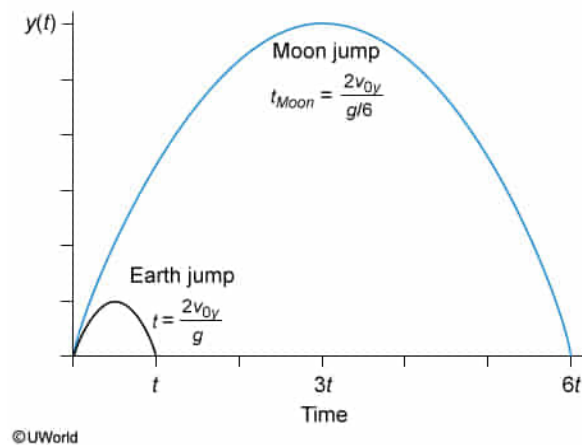
Provided that the initial velocity in both circumstances is identical, how much longer will an astronaut's vertical jumps last on the surface of the Moon compared to on the surface of the Earth?

- ☐ A. 1/6 as long [4%]
- ☐ B. 1/3 as long [1%]
- ☐ C. 3 times longer [13%]
- ☒ D. 6 times longer [80%]

Correct

80% answered correctly

Explanation:



The **time of flight** for projectiles can be determined from [translation motion equations](#). Specifically, the linear position x of an object with constant acceleration a is given by the kinematic equation:

$$x = x_0 + vt + \frac{1}{2}at^2$$

In the case of vertical motion (y -axis), the acceleration of gravity g points downward and is therefore negative:

$$y = y_0 + v_y t + -\frac{1}{2}gt^2$$

When an astronaut completes a vertical jump, the difference between the final and initial position ($y - y_0$) is 0:

$$y - y_0 = v_y t - \frac{1}{2}gt^2$$

$$0 = t \left(-\frac{1}{2}gt + v_y \right)$$

The equation above has two solutions when rearranged to solve for the time t of the jump. The first solution is $t = 0$, corresponding to the beginning of the jump. The second solution is when the expression in parenthesis is equal to 0:

$$-\frac{1}{2}gt + v_y = 0$$

$$-\frac{1}{2}gt = -v_y$$

$$t = -\frac{-2v_y}{g}$$

$$t = \frac{2v_y}{g}$$

Hence, t is **inversely proportional** to g . Modifying the above equation to account for the significantly reduced g on the Moon (g_{Moon}) yields:

$$t_{\text{Moon}} = \frac{2v_y}{g_{\text{Moon}}} = \frac{2v_y}{g/6} = 6\frac{2v_y}{g} = 6t$$

Therefore, an object launched vertically from the surface of the Moon would take **6 times longer** to return than an object launched vertically from the surface of the Earth (**Choice D**).

Educational objective:

Translational motion equations can be used to determine the height and travel time of a projectile, the latter of which is inversely proportional to the acceleration of gravity.

Time spent:0 sec

Subject:
Physics

QID:400630

System:
Mechanics and Energy